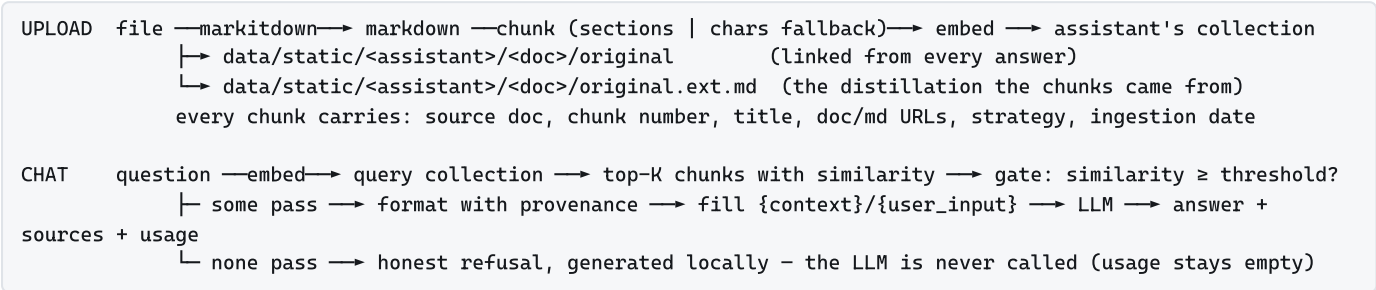


W2-Final — EASY-RAG

Exercise 1's EASY-ASSISTANT grown into a real RAG app: FastAPI backend, plain-JS frontend, documents ingested into a per-assistant collection through the course's `collections-manager` (vendored as a path dependency — the app never talks to ChromaDB directly), answers grounded in retrieved chunks with visible provenance. Models: `qwen3:1.7b` answers (run locally as `qwen3-fast` — the same weights pinned to 6 CPU threads), `nomic-embed-text` embeds, both on local Ollama.

HOW INGESTION AND RETRIEVAL FLOW



THE THREE KNOBS, AND WHY

Knob	Value	Why
chunking	<code>sections</code> (level 2), falling back to <code>chars</code> 800/100	In exercise 3 I ingested the same document both ways: section chunks carry their own heading and one complete idea, retrieved the exact right section every time (0.675 vs 0.623 on the same question) and kept the model from inventing facts out of window fragments — the fallback covers documents that convert without headings.
K	4	Enough for an answer that spans a couple of sections, while keeping the prompt small for a 1.7B model — with section-sized chunks, K=4 already means ~2k prompt tokens.
threshold	0.57	Calibrated in exercise 3 against <code>nomic-embed-text</code> and my documents: off-topic questions peaked at 0.552 (not near zero!) and real questions started at 0.608, so 0.57 sits inside the narrow band that blocks noise without starving real answers.

All three are `.env` configuration, not constants — the threshold in particular is a calibration that belongs to this embeddings model and corpus, not a universal number.

GROUNDING ANSWER, WITH ITS PROVENANCE

EASY-RAG

qwen3-fast · nomic-embed-text · top-4 ≥ 0.57 · sections

ASSISTANTS

New

Trebol Docs

4 docs · 56 chunks

Trebol Docs

4 docs · 56 chunks

Trebol Docs

4 docs · 56 chunks

Upload document

Edit

Delete

architecture.md

17 chunks · sections · 26.5k chars

md

x

DESIGN.md

16 chunks · sections · 61.6k chars

md

x

tortuga-robotics.txt

5 chunks · sections · 3.2k chars

md

x

ex_01_report.pdf

18 chunks · chars · 12.6k chars

md

x

YOU

How fast is the delivery robot?

TREBOL DOCS

The delivery robot (TR-100 "Galápagó") has a maximum speed of **1.1 metres per second**.

answered from:

tortuga-robotics · #1 · 0.646

tortuga-robotics · #2 · 0.583

tortuga-robotics · #0 · 0.572

Ask something about the documents... (Enter to send, Shift+Enter for a newline)

Send

Streaming

New chat

Context

RETRIEVAL

The question became the query. ✓ passed the threshold and reached the prompt; ✗ was retrieved but rejected. When every row is ✗, the assistant refuses instead of answering from noise.

top-4 over 0.57 — 3 injected

✓ 0.646

tortuga-robotics · chunk 1

✓ 0.583

tortuga-robotics · chunk 2

✓ 0.572

tortuga-robotics · chunk 0

✗ 0.536

tortuga-robotics · chunk 3

FILLED PROMPT

The template with `{context}` replaced by the retrieved chunks and `{user_input}` by your question. This is the user message that went upstream on the last turn.

Use only the information in the context below to answer the question.
If the answer is not in the context, say that you do not know.

Context:

[tortuga-robotics · chunk 1 · similarity 0.646]
Ferrer-Solé.

PRODUCTS
The TR-100 "Galápagó" is the flagship. It is a floor transport robot that carries up to 240 kg at a maximum speed of 1.1

FULL PAYLOAD

Every message the backend sent, system prompt included.

SYSTEM

You are a careful assistant that answers strictly from the context you are given. If the context does not contain the answer, say that you do not know. You never invent facts.

USER

Four documents in one assistant (56 chunks: a microservice architecture guide, a design system, a robotics fact sheet, and a PDF ingested through the `chars` fallback). *"How fast is the delivery robot?"* retrieves chunks only from the right document; the chips under the answer link to the original under `/static`, and the panel shows each chunk's similarity and whether it passed the gate (✓ injected, ✗ dropped at 0.536 < 0.57).

HONEST REFUSAL

EASY-RAG

qwen3-fast · nomic-embed-text · top-4 ≥ 0.57 · sections

Streaming

New chat

Context

ASSISTANTS

New

Trebol Docs

4 docs · 56 chunks

Upload document · Edit · Delete

architecture.md 17 chunks · sections · 26.5k chars · md · x · DESIGN.md 16 chunks · sections · 61.6k chars · md · x · tortuga-robotics.txt 5 chunks · sections · 3.2k chars · md · x · ex_01_report.pdf 18 chunks · chars · 12.6k chars · md · x

YOU

What is the recipe for paella valenciana?

TREBOL DOCS

I don't know — nothing in this assistant's documents is close enough to your question to answer from. Try rephrasing, or upload a document that covers it.

.....

nothing passed the threshold (best 0.514 < 0.57)

Ask something about the documents... (Enter to send, Shift+Enter for a newline)

Send

RETRIEVAL

The question became the query. ✓ passed the threshold and reached the prompt; ✗ was retrieved but rejected. When every row is ✗, the assistant refuses instead of answering from noise.

top-4 over 0.57 — REFUSED, nothing injected

✗ 0.514 DESIGN · chunk 0

✗ 0.510 architecture · chunk 14

✗ 0.493 architecture · chunk 2

✗ 0.488 architecture · chunk 4

FILLED PROMPT

The template with {context} replaced by the retrieved chunks and {user_input} by your question. This is the user message that went upstream on the last turn.

—

FULL PAYLOAD

Every message the backend sent, system prompt included.

TOKENS

Prompt

Completion

Last turn total

Session total

0

Watch Prompt: it no longer grows with the size of your documents — only the retrieved chunks ride along, however big the collection gets. Compare with exercise 1.

"What is the recipe for paella valenciana?" — the best chunk reaches only 0.514, every row shows X, nothing is injected, and the assistant refuses. The Tokens panel stays empty: no request went upstream, which is what makes the refusal honest.

PROMPT_TOKENS, COMPARED WITH EXERCISE 1

In exercise 1 my `prompt_tokens` went **968 → 1,007** turn over turn with a single 3,193-character document — the whole file rode along every time, so the prompt could only grow with the knowledge. EASY-RAG holds a **104,000-character corpus** (4 documents, 56 chunks) and the same questions cost **372–827** prompt tokens — and when the corpus grew from 3 documents to 4, those numbers did not move by a single token, because only the retrieved chunks travel: **the prompt scales with the question, not with the corpus** — a corpus ~32× larger than exercise 1's document produced prompts of the same size or smaller, and that is the reason this week existed.